gpeyt1-29 · Confined aquifer K

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A pumping test in a 30 ft thick confined sand aquifer (overlain by 22 ft of clay) runs at a steady 350 gpm. Drawdowns: 9.2 ft at r = 40 ft and 5.1 ft at r = 200 ft. The pumping well radius is 0.67 ft; the piezometric surface is 8 ft below grade before pumping.

Most nearly, the hydraulic conductivity K is:

A. 140 ft/day
B. 61 ft/day
C. 40 ft/day
D. 281 ft/day

Given equations / data

> Confined: $Q = 2\pi Kb(h_2 - h_1)/\ln(r_2/r_1)$; $h_2 - h_1 = s_1 - s_2$; 1 ft^3 = 7.48 gal

Worked solution

1. $Q = 350 \text{ gpm} / 7.48 \times 1{,}440 = 67{,}380 \text{ ft}^3\text{/day}$
2. $h_2 - h_1 = 9.2 - 5.1 = 4.1 \text{ ft}$
3. $K = 67{,}380 \times \ln(200/40) / (2\pi \times 30 \times 4.1) = 140.3 \text{ ft/day}$

Final answer

140 ft/day

Why the other values are wrong

- **61 ft/day** — used log10
- **40 ft/day** — added drawdowns
- **281 ft/day** — used π instead of 2π

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §6.6.4 Well Analysis–Steady State (confined aquifer); UFC 3-220-05 Ch. 4

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

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